1 I

1.1 (1)

From the definitions of the creation and annihilation operators,

$$a = \sqrt{\frac{m\omega}{2\hbar}}\left(x + \frac{ip}{m\omega}\right), \qquad a^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(x - \frac{ip}{m\omega}\right),$$

we obtain

$$\begin{aligned}
[a, a^\dagger] &= aa^\dagger - a^\dagger a \\
&= \frac{i}{2\hbar}([p, x] - [x, p]) \\
&= \frac{i}{\hbar}[p, x] \\
&= 1.
\end{aligned}$$

1.2 (2)

Using $[a, a^\dagger] = 1$, we have

$$[a, (a^\dagger)^n] = n(a^\dagger)^{n-1}.$$

Therefore,

$$\begin{aligned}
[H, (a^\dagger)^n] &= \hbar\omega[a^\dagger a, (a^\dagger)^n] \\
&= \hbar\omega a^\dagger[a, (a^\dagger)^n] \\
&= n\hbar\omega(a^\dagger)^n.
\end{aligned}$$

Since $a|0\rangle = 0$, the ground-state energy satisfies

$$\begin{aligned}
H|0\rangle &= \hbar\omega\left(a^\dagger a + \frac{1}{2}\right)|0\rangle \\
&= \frac{1}{2}\hbar\omega|0\rangle = E_0|0\rangle.
\end{aligned}$$

For $|n\rangle = (a^\dagger)^n|0\rangle$, we obtain

$$\begin{aligned}
H|n\rangle &= H(a^\dagger)^n|0\rangle \\
&= \left([H, (a^\dagger)^n] + (a^\dagger)^n H\right)|0\rangle \\
&= (n\hbar\omega + E_0)|n\rangle \\
&= \hbar\omega\left(n + \frac{1}{2}\right)|n\rangle.
\end{aligned}$$

Hence, $|n\rangle$ is an eigenstate of H with eigenvalue

$$\boxed{E_n = \hbar\omega\left(n + \frac{1}{2}\right)}.$$

2 II

2.1 (3)

Define the Hamiltonian along each coordinate axis as

$$H_i = \frac{p_i^2}{2m} + \frac{m\omega^2 x_i^2}{2}.$$

The total Hamiltonian is

$$H = \sum_{i=1}^{3} H_i.$$

Since $[x_i, p_j] = i\hbar\delta_{ij}$ and operators associated with different coordinate axes commute, we have

$$[H_i, H_j] = 0.$$

Thus, the three-dimensional harmonic oscillator can be treated as three independent one-dimensional oscillators:

$$H_i|n_i\rangle_i = \hbar\omega\left(n_i + \frac{1}{2}\right)|n_i\rangle_i.$$

The eigenstates of the total Hamiltonian are

$$|n_1, n_2, n_3\rangle = (a_1^\dagger)^{n_1}(a_2^\dagger)^{n_2}(a_3^\dagger)^{n_3}|0\rangle,$$

where normalization factors have been omitted.

Therefore,

$$\begin{aligned} H|n_1, n_2, n_3\rangle &= \sum_{i=1}^{3} H_i|n_1, n_2, n_3\rangle \\ &= \hbar\omega\left(n_1 + n_2 + n_3 + \frac{3}{2}\right)|n_1, n_2, n_3\rangle. \end{aligned}$$

Hence, the energy eigenvalues are

$$\boxed{E_N = \hbar\omega\left(N + \frac{3}{2}\right), \qquad N = n_1 + n_2 + n_3.}$$

2.2 (4)

The position and momentum operators can be expressed as

$$\begin{aligned} x_i &= \sqrt{\frac{\hbar}{2m\omega}}(a_i + a_i^\dagger), \\ p_i &= i\sqrt{\frac{\hbar m\omega}{2}}(a_i^\dagger - a_i). \end{aligned}$$

Substituting them into the orbital angular momentum operator,

$$L_i = \sum_{j,k=1}^{3} \epsilon_{ijk} x_j p_k,$$

we obtain

$$\begin{aligned} L_i &= \frac{i\hbar}{2}\sum_{j,k=1}^{3}\epsilon_{ijk}(a_j + a_j^\dagger)(a_k^\dagger - a_k) \\ &= \frac{i\hbar}{2}\sum_{j,k=1}^{3}\epsilon_{ijk}(a_j a_k^\dagger - a_j^\dagger a_k) \\ &= i\hbar\sum_{j,k=1}^{3}\epsilon_{ijk}a_j a_k^\dagger \\ &= -i\hbar\sum_{j,k=1}^{3}\epsilon_{ijk}a_j^\dagger a_k. \end{aligned}$$

Here, we have used the antisymmetry of ϵ_{ijk} and $[a_j, a_k^\dagger] = \delta_{jk}$.

Since $a_k|0\rangle = 0$ for all k,

$$\begin{aligned} L_i|0\rangle &= -i\hbar\sum_{j,k=1}^{3}\epsilon_{ijk}a_j^\dagger a_k|0\rangle \\ &= 0. \end{aligned}$$

Consequently,

$$L_i^2|0\rangle = 0, \qquad L^2|0\rangle = \sum_{i=1}^{3}L_i^2|0\rangle = 0.$$

Comparing this with

$$L^2|0\rangle = \hbar^2 l(l+1)|0\rangle,$$

we conclude that the ground state has

$$\boxed{l = 0}.$$

2.3 (5)

Using the expression for L_i obtained in (4),

$$\begin{aligned} [L_i, a_j^\dagger] &= -i\hbar\sum_{k,l=1}^{3}\epsilon_{ikl}[a_k^\dagger a_l, a_j^\dagger] \\ &= -i\hbar\sum_{k,l=1}^{3}\epsilon_{ikl}a_k^\dagger[a_l, a_j^\dagger] \\ &= -i\hbar\sum_{k=1}^{3}\epsilon_{ikj}a_k^\dagger \\ &= i\hbar\sum_{k=1}^{3}\epsilon_{ijk}a_k^\dagger. \end{aligned}$$

Therefore,

$$\boxed{[L_i, a_j^\dagger] = i\hbar\sum_{k=1}^{3}\epsilon_{ijk}a_k^\dagger}.$$

For the double commutator,

$$\begin{aligned}\sum_{i=1}^{3}[L_i,[L_i,a_j^\dagger]] &= i\hbar\sum_{i,k=1}^{3}\epsilon_{ijk}[L_i,a_k^\dagger]\\ &= -\hbar^2\sum_{i,k,l=1}^{3}\epsilon_{ijk}\epsilon_{ikl}a_l^\dagger\\ &= 2\hbar^2 a_j^\dagger,\end{aligned}$$

where the identity

$$\sum_{i,k=1}^{3}\epsilon_{ijk}\epsilon_{ikl} = -2\delta_{jl}$$

has been used.

Hence,

$$\boxed{\sum_{i=1}^{3}[L_i,[L_i,a_j^\dagger]] = 2\hbar^2 a_j^\dagger}.$$

2.4 (6)

The first excited energy level E_1 is threefold degenerate, with eigenstates

$$|1,0,0\rangle, \qquad |0,1,0\rangle, \qquad |0,0,1\rangle.$$

Each of these states can be expressed as

$$a_j^\dagger|0\rangle, \qquad j=1,2,3.$$

Since $L_i|0\rangle = 0$, we have

$$\begin{aligned}L_i a_j^\dagger|0\rangle &= ([L_i,a_j^\dagger] + a_j^\dagger L_i)|0\rangle\\ &= [L_i,a_j^\dagger]|0\rangle.\end{aligned}$$

Applying L_i once more,

$$L_i^2 a_j^\dagger|0\rangle = [L_i,[L_i,a_j^\dagger]]|0\rangle.$$

Summing over i and using the result from (5),

$$\begin{aligned}L^2 a_j^\dagger|0\rangle &= \sum_{i=1}^{3}[L_i,[L_i,a_j^\dagger]]|0\rangle\\ &= 2\hbar^2 a_j^\dagger|0\rangle.\end{aligned}$$

Comparing this with

$$L^2|\psi\rangle = \hbar^2 l(l+1)|\psi\rangle,$$

we obtain

$$l(l+1) = 2, \qquad \boxed{l=1}.$$

Thus, all states in the first excited energy level have orbital angular momentum $l=1$.

3 III

3.1 (7)

Since the particles are identical spin-1/2 fermions, the total two-particle wavefunction must be antisymmetric under particle exchange.

Ground State Both particles occupy the lowest spatial orbital $|0,0,0\rangle$, whose energy is

$$E_0 = \frac{3}{2}\hbar\omega.$$

Since their spatial wavefunction is symmetric, their spin wavefunction must be antisymmetric, corresponding to the spin-singlet state

$$\chi_{0,0} = \frac{1}{\sqrt{2}}\left(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle\right).$$

Therefore, the ground state is

$$|\Psi_{\mathrm{G}}\rangle = |0,0,0\rangle_1|0,0,0\rangle_2\,\chi_{0,0},$$

with total energy

$$\boxed{E_{\mathrm{G}} = 3\hbar\omega}.$$

The ground state is nondegenerate.

First Excited States In the first excited configuration, one particle remains in the ground spatial orbital, while the other occupies one of the three first excited orbitals:

$$|1,0,0\rangle, \qquad |0,1,0\rangle, \qquad |0,0,1\rangle.$$

Let $|g\rangle = |0,0,0\rangle$ and let $|e_j\rangle$ denote one of these three excited orbitals.

The two-particle spatial wavefunctions can be constructed as

$$|\psi_{\pm}^{(j)}\rangle = \frac{1}{\sqrt{2}}\left(|g\rangle_1|e_j\rangle_2 \pm |e_j\rangle_1|g\rangle_2\right).$$

For identical fermions, the total wavefunction must be antisymmetric. Thus, there are two possibilities:

- A symmetric spatial wavefunction $|\psi_{+}^{(j)}\rangle$ combined with the antisymmetric spin singlet ($S = 0$).
- An antisymmetric spatial wavefunction $|\psi_{-}^{(j)}\rangle$ combined with a symmetric spin triplet ($S = 1$).

The spin-triplet states are

$$\begin{aligned}
\chi_{1,1} &= |\uparrow\uparrow\rangle, \\
\chi_{1,0} &= \frac{1}{\sqrt{2}}\left(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle\right), \\
\chi_{1,-1} &= |\downarrow\downarrow\rangle.
\end{aligned}$$

The energy of each first excited configuration is

$$\begin{aligned} E_{\mathrm{F}} &= \frac{3}{2}\hbar\omega + \frac{5}{2}\hbar\omega \\ &= \boxed{4\hbar\omega}. \end{aligned}$$

For each of the three excited spatial orbitals, there is one spin-singlet state and three spin-triplet states. Therefore, the first excited energy level has

$$\boxed{3(1+3) = 12}$$

degenerate states.

3.2 (8)

Ground State Both particles occupy spatial orbitals with $l = 0$, so their total orbital angular momentum is

$$L = 0.$$

Since their spin state is a singlet,

$$S = 0.$$

Therefore,

$$\boxed{J = 0}.$$

First Excited States One particle has orbital angular momentum $l_1 = 0$, while the other has $l_2 = 1$. Therefore, the total orbital angular momentum is

$$L = 1.$$

For the spin-singlet states ($S = 0$), the only possible total angular momentum is

$$J = L = 1.$$

For the spin-triplet states ($S = 1$), angular momentum addition gives

$$J = |L - S|, \ldots, L + S,$$

so

$$J = 0, 1, 2.$$

Combining these possibilities, the allowed total angular momenta for the first excited states are

$$\boxed{J = 0, 1, 2}.$$